INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

(ICDFA)

Basic Computer Skills for Digital Forensics

LAB 1B

PC System Fundamentals, Disk Sectors and Partition Analysis

PRACTICAL LABORATORY REPORT

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1. Introduction

This report presents the findings of a read-only forensic profile carried out on a personal workstation for Lab 1B of CIP-B101. The computer examined is an HP ZBook 15u G3 running Microsoft Windows 10 Pro (version 10.0.19045, build 19045) with an Intel Core i7-6500U processor, 16 GB of RAM and two internal solid-state drives: a 240 GB Intel SATA SSD and a 256 GB SK hynix NVMe drive. The purpose of the exercise was to identify the hardware and software characteristics of the system, map the layers from which digital evidence can be generated, observe running processes, inspect the partition table, calculate partition sizes from raw sector counts, explain the boot sequence, compare MBR with GPT, and identify the file systems in use. Every activity was performed with read-only tools, and no partition, file system or disk structure was modified at any point.

1.1 Safety and evidence-handling compliance

The lab was treated as an observation-only exercise. The commands below are all read-only, and the following were deliberately never executed: format, clean, create partition, delete partition, convert, active, shrink, extend, mkfs, dd or any interactive fdisk/parted write operation. In DiskPart only list disk, select disk 0, list partition and exit were used; select does not alter the disk.

- msinfo32 and systeminfo – display system and operating system information.
- wmic diskdrive / wmic logicaldisk get … – display storage and volume information.
- PowerShell: Get-Disk, Get-Partition, Get-Volume, Get-CimInstance Win32_DiskPartition – read-only queries.
- Disk Management (diskmgmt.msc) – visual inspection of the disk layout only.
- DiskPart – listing commands only (list disk, list partition).
- Task Manager and tasklist – observation of running processes.

Serial numbers appear in some screenshots because they were captured directly from the live machine. They are not repeated in the body text, and the report contains no usernames, e-mail addresses or account credentials. Partition sizes, offsets and file systems were left fully visible because they constitute the graded evidence.

2. Tools and environment

Tool / command	Purpose in this lab	Read-only?
Windows 10 Pro 22H2 (build 19045), x64	Host operating system used for all observations	–
System Information (msinfo32)	System summary, BIOS mode, memory, processor	Yes
systeminfo (Command Prompt)	Operating system and hardware summary	Yes
wmic diskdrive / wmic logicaldisk	Disk model, size, serial; volume file system and free space	Yes
PowerShell 5.1: Get-Disk, Get-Partition, Get-Volume	Disk, partition and volume details	Yes
Get-CimInstance Win32_DiskPartition	Starting offset, sector count and block size per partition	Yes
Disk Management (diskmgmt.msc)	Visual partition layout and partition style (MBR/GPT)	Yes
DiskPart (list commands only)	Independent partition-table listing with offsets in KB	Yes
Task Manager (Details tab) and tasklist	Running processes and PIDs	Yes
Python 3.14	Arithmetic verification of all sector calculations	Yes
Windows Snipping Tool	Evidence screenshots	Yes

3. Part A – System identification and evidence-layer mapping

3.1 System specification (A1)

System Information was opened with msinfo32 and the System Summary captured. The key values are summarised below; the screenshot itself is Figure 1.

Item	Observed value
OS Name	Microsoft Windows 10 Pro
Version / Build	10.0.19045 Build 19045 (64-bit, x64-based PC)
System Manufacturer / Model	HP / HP ZBook 15u G3 (System SKU M6G49AV)
Processor	Intel(R) Core(TM) i7-6500U CPU @ 2.50 GHz, 2 cores / 4 logical processors, 2592 MHz
BIOS Version/Date	HP N75 Ver. 01.42, 19/08/2019; SMBIOS 2.7
BIOS Mode	UEFI
Secure Boot State	Elevation Required to View (not readable without administrator rights)
Installed Physical Memory	16.0 GB (17,054,666,752 bytes ≈ 15.88 GiB)
Boot Device	\Device\HarddiskVolume5
Time Zone	(UTC+01:00) W. Central Africa Standard Time

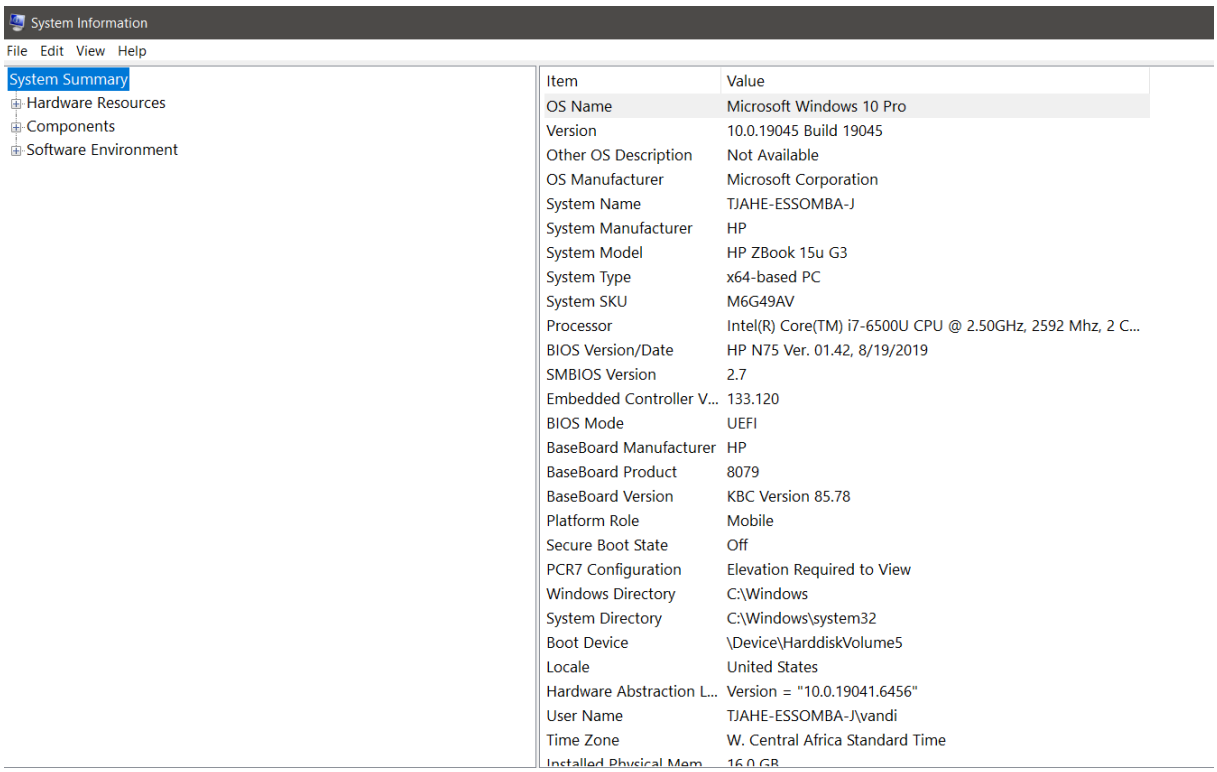


Figure 1 – msinfo32 System Summary: Windows 10 Pro, i7-6500U, 16 GB RAM, BIOS Mode UEFI (screenshot 1).

The most important observation for this lab is the BIOS Mode: UEFI. A UEFI system normally boots from a GPT-partitioned disk, and this is confirmed later by Get-Disk, which shows the system/boot NVMe disk as GPT. Secure Boot state could not be read because msinfo32 requires elevation for that field; this is a limitation of the read-only, non-elevated session, not a fault of the machine. The firmware is from 2019, which fits the 2016-era Skylake platform of the ZBook.

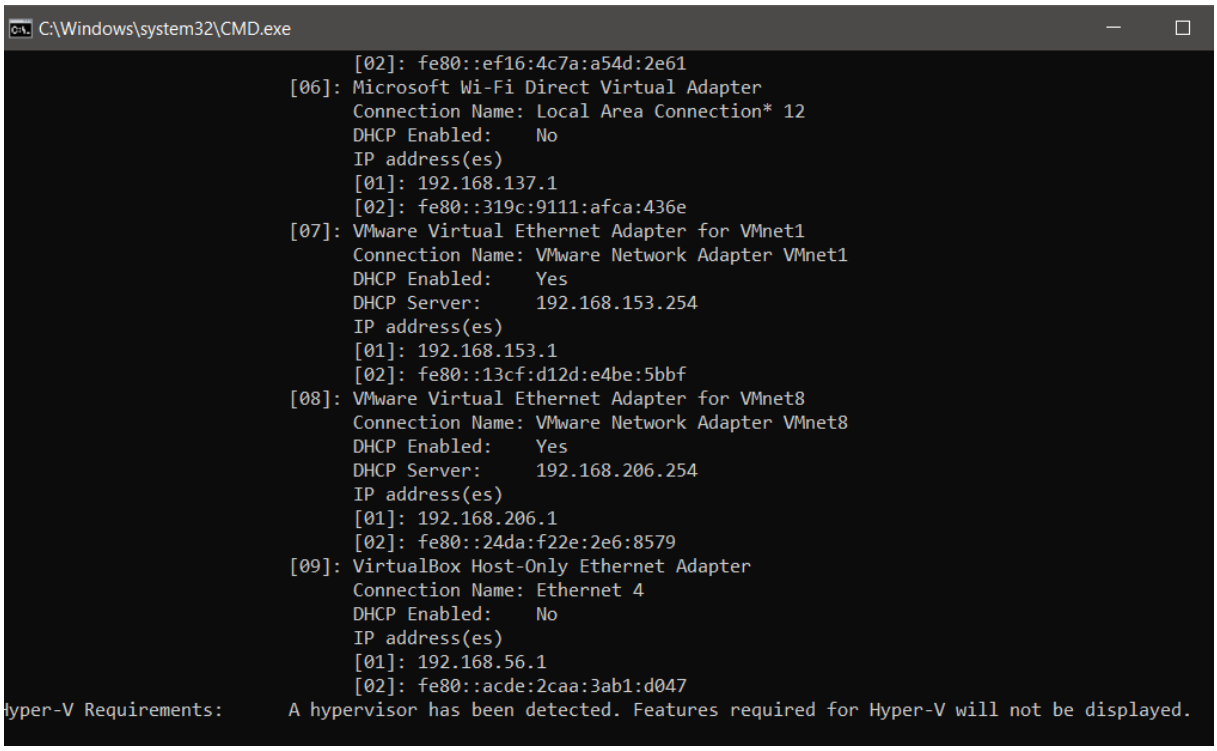


Figure 2 – systeminfo output (network adapters and Hyper-V requirements section) (screenshot 2).

The systeminfo output confirms the same platform information and also lists several virtual network adapters (VMware VMnet1/VMnet8, VirtualBox Host-Only, Hyper-V). This is relevant forensically: virtual network interfaces show that virtualisation software is installed, which could mean virtual machines and virtual disks exist on the system and should be considered during evidence collection.

3.2 Storage information (A2)

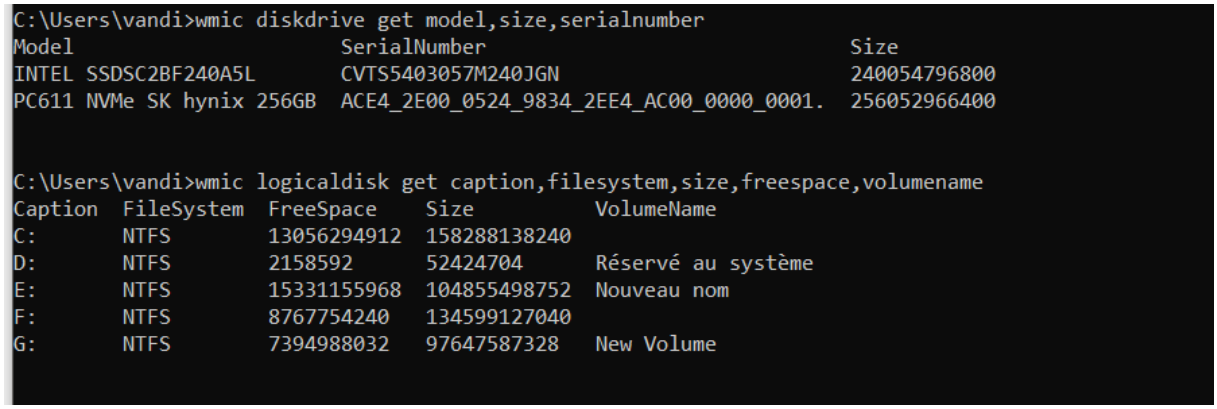


Figure 3 – wmic diskdrive and wmic logicaldisk output: disk models/sizes and volume file systems (screenshot 3).

WMIC is deprecated but still functional on build 19045, so both the WMIC output and the PowerShell output could be collected. The machine has two physical disks, summarised below. Serial numbers are visible in the screenshot but are intentionally omitted from the table.

Disk	Model	Raw capacity	Interface	Partition style	Role
0	INTEL SSDSC2BF240A5L	240,055,312,384 B (223.57 GiB / 240 GB class)	SATA SSD	MBR	Data disk
1	PC611 NVMe SK hynix 256GB	256,059,465,728 B (238.47 GiB / 256 GB class)	NVMe	GPT	System / boot disk

The difference between the advertised capacity and the Windows capacity is a units effect, not missing space: 240,055,312,384 bytes ÷ 1,000,000,000 = 240.06 decimal GB, but ÷ 1,073,741,824 = 223.57 GiB. Windows labels binary GiB values as "GB". The same applies to the 256 GB NVMe drive (238.47 GiB).

Volume	File system	Size	Free space (at capture)	Location / note
C:	NTFS	147.42 GB	11.78 GB	Disk 1, Partition 3 – Windows system volume
G:	NTFS	90.94 GB	6.89 GB	Disk 1, Partition 4 – data volume
F:	NTFS	125.36 GB	8.17 GB	Disk 0, Partition 2 – data volume
E:	NTFS	97.65 GB	14.28 GB	Disk 0 – logical drive inside the extended partition
D:	NTFS	50 MB	2.06 MB	Disk 0, Partition 1 – "Réservé au système"
(no letter)	NTFS	519 MB	84.71 MB	Disk 0, Partition 3 – recovery partition
(no letter)	FAT32	100 MB	–	Disk 1, Partition 1 – EFI System Partition
(no letter)	None (reserved)	16 MB	–	Disk 1, Partition 2 – Microsoft Reserved (MSR)

3.3 Evidence sources mapped to system layers (A3)

Layer / source	Example evidence on this machine	Where it may be found	Why it matters
User layer	Recently opened files, browser history and searches, typed usernames, clipboard content	NTFS Recent items, Chrome/Brave History databases, PowerShell console history	Links a specific person to actions performed on the machine
Application layer	Office document metadata (author, last saved by), browser cache, chat and e-mail stores	AppData folders, application SQLite databases, Office document properties	Shows which software created or used the data and can retain copies of deleted content
Operating system layer	Event logs, registry hives, Prefetch files, installed programs, user profiles	C:\Windows\System32\winevt\Logs, NTUSER.DAT, C:\Windows\Prefetch	Records system events, program execution and account activity
Disk layer	Deleted files, file fragments, partition table, slack space, unallocated clusters	NTFS $MFT, $LogFile, unallocated space, MBR/GPT structures	Enables file recovery, timeline analysis and verification of disk structure
Memory / process layer	Running processes and PIDs, network connections, injected code, encryption keys	RAM capture, tasklist/Get-Process, netstat output	Shows live behaviour that disappears when the machine is shut down (volatile evidence)
Network layer	DNS cache, ARP table, active connections, browser sessions, firewall logs	ipconfig /displaydns, arp -a, netstat -ano, firewall and proxy logs	Shows remote interaction and can link the host to external systems

Evidence can exist in more than one layer because a single user action passes through all of them. For example, when a user opens a document, the application records it in a recent-files list, the operating system logs the event and updates registry keys, the file content is stored on disk, and the data is loaded into RAM while the document is open. Each layer therefore holds a partial, sometimes independent, copy of the same event. This redundancy is what allows an investigator to cross-verify a finding: if the user claims a file was never opened, the application history, the registry and the process/memory evidence can be compared against that statement.

4. Part B – Program, process and thread

4.1 Observed running processes (B1)

Task Manager was opened on the Details tab, where the PID column is available, and tasklist was run in Command Prompt as a second, text-based source.

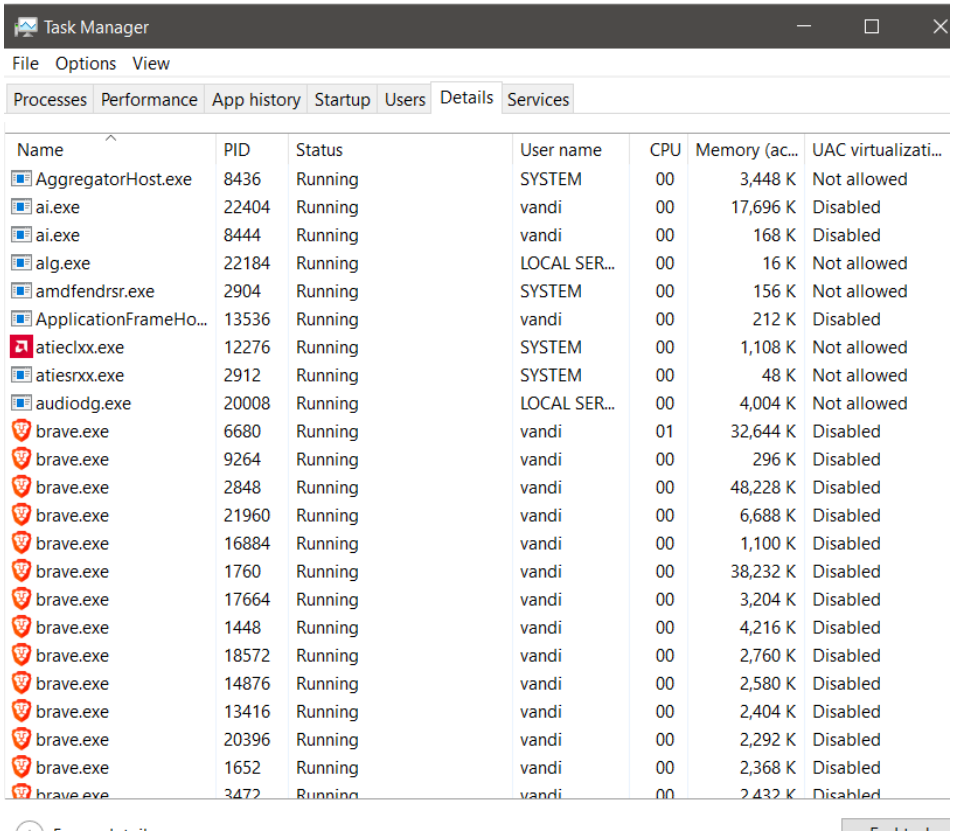


Figure 4 – Task Manager Details tab showing process names, PIDs and user accounts (screenshot 4).

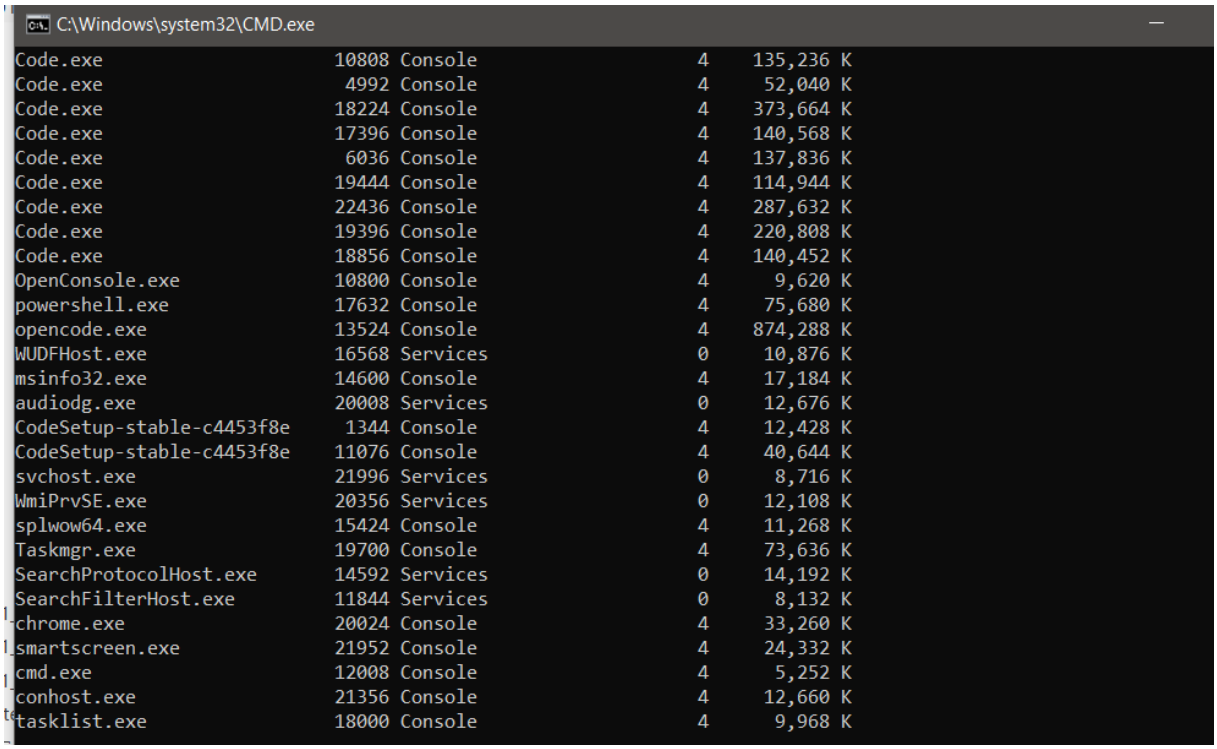


Figure 5 – tasklist output in Command Prompt (screenshot 5).

Three processes were selected from the captured evidence. The PIDs below are the values visible in Figures 4 and 5 at the time of capture; PIDs are assigned at process creation and change after a reboot.

Process name	PID	User context	Likely purpose
AggregatorHost.exe	8436	SYSTEM	Windows security component used by Microsoft Defender/telemetry to aggregate scan, health and reporting data. It runs as SYSTEM and has no visible window.

Process name	PID	User context	Likely purpose
audiodg.exe	20008	LOCAL SERVICE	Audio Device Graph Isolation. Hosts the audio engine that processes sound for applications; it is kept in a separate process for stability and security.
brave.exe	6680	vandi (interactive user)	Main process of the Brave web browser. It creates network connections and spawns child processes for tabs and renderers; several brave.exe entries are visible for this reason.

Two observations are worth recording. First, the machine shows many svchost.exe instances, which is normal because Windows groups services into shared service-host processes. Second, both chrome.exe and brave.exe appear many times; modern Chromium browsers use a multi-process architecture in which each tab, GPU task and utility runs in its own process, and each process can contain many threads. This is exactly why a process list alone is not enough during triage: the analyst must also look at parent processes, paths, signatures and network connections.

4.2 Program, process and thread explained (B2)

A program is a passive file stored on disk that contains instructions, such as notepad.exe or brave.exe. It does nothing until it is executed. A process is an instance of a program that has been loaded into memory and is running; it has its own PID, virtual address space, handles and security context. A thread is the smallest unit of execution inside a process; threads of the same process share its memory and resources, which makes them cheap to create but also means a fault in one thread can affect the whole process.

In forensic terms, the process list is volatile evidence. It can show that malware is currently running, reveal suspicious names, unexpected parent–child relationships, unusual user contexts or network connections, and it provides the PIDs needed to dump a process memory image. Because all of this disappears when the machine is powered off, live triage must capture it before shutdown. A program on disk, by contrast, survives reboots and can be hashed and analysed later; the two sources complement each other.

5. Part C – Disk sector and disk size calculations

The fundamental relationship used throughout this lab is: size in bytes = number of sectors × bytes per sector. To convert bytes to GB (binary, as used by Windows and by this lab), the value is divided by 1024 three times.

5.1 Reproduction of the slide calculation (C1)

Given: sectors per track = 400, heads = 12, cylinders = 17,000 and bytes per sector = 512.

Step	Working	Result
Total sectors	400 × 12 × 17,000	81,600,000 sectors
Total bytes	81,600,000 × 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 ÷ 1,024	40,800,000 KB
Megabytes	40,800,000 ÷ 1,024	39,843.75 MB
Gigabytes (GiB)	39,843.75 ÷ 1,024	38.9099 GB ≈ 38.91 GB

```
>>> 400 * 12 * 17000 * 512 / 1024**3
38.909912109375
```

The Python check returns the same value, 38.9099 GiB, which rounds to the 38.91 GB shown on the slide. The CHS (cylinder–head–sector) addressing used here is the classic geometry method; modern drives are addressed by LBA, but the arithmetic of sectors × sector size is unchanged.

5.2 Size of the two disks in this PC (C2)

The capacity reported by Get-Disk was converted back into sectors to confirm the arithmetic:

```
Disk 0: 240,055,312,384 bytes ÷ 512 = 468,858,032 sectors  (223.57 GiB)
Disk 1: 256,059,465,728 bytes ÷ 512 = 500,116,144 sectors  (238.47 GiB)
```

The two partitions selected for detailed analysis are Partition 1 and Partition 2 of Disk 0, as listed by Get-Partition and DiskPart. The full step-by-step calculation is given in Section 9 (Part G) so that the verification process and the calculation appear together, as required by the assignment. The results are: Partition 1 = 102,400 sectors = 52,428,800 bytes = 50 MB = 0.0488 GiB; Partition 2 = 262,888,926 sectors = 134,599,130,112 bytes = 125.3552 GiB ≈ 125.36 GB.

6. Part D – Safe partition-table inspection

6.1 Disk Management (D1)

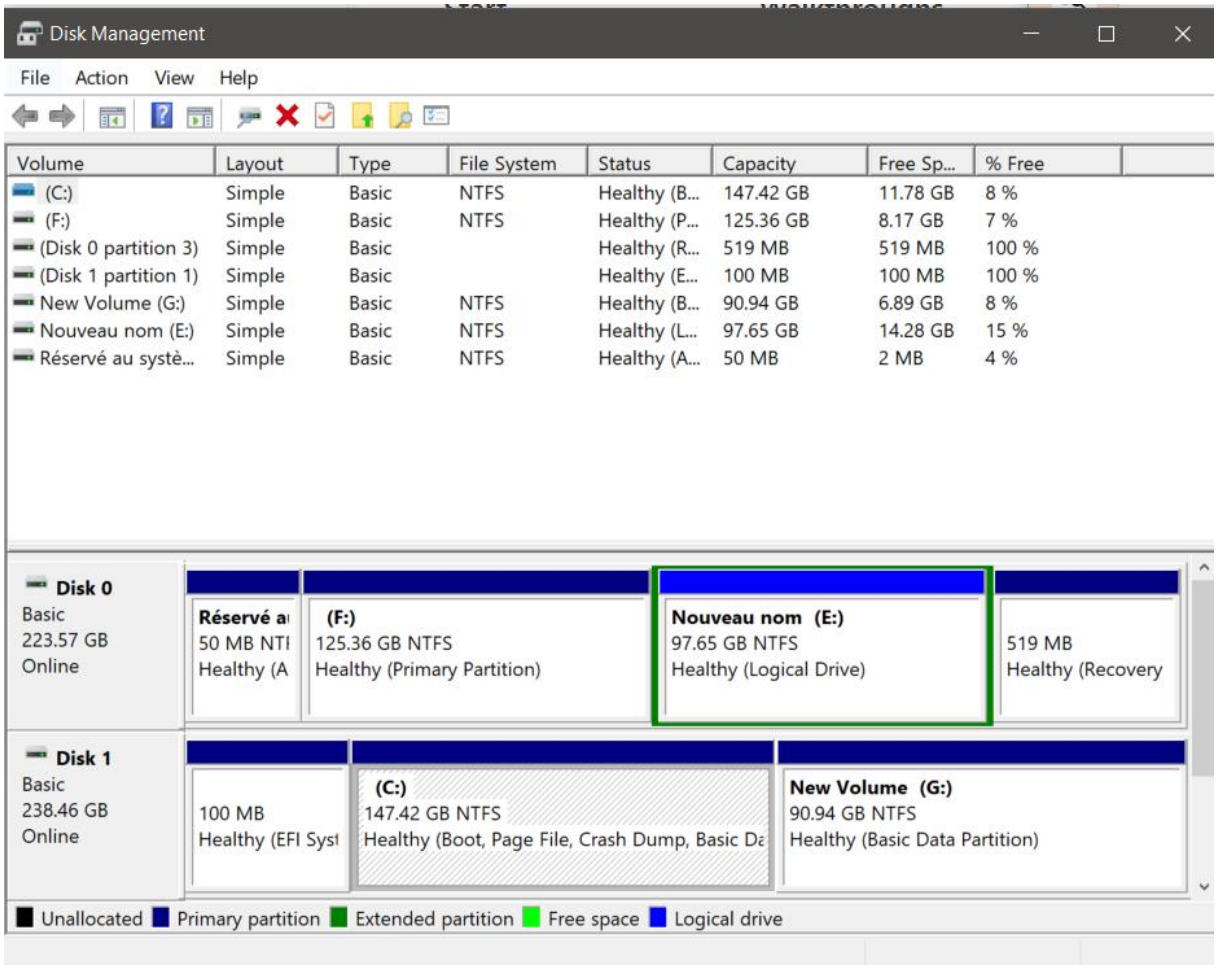


Figure 6 – Disk Management showing both disks and all visible partitions (screenshot 6).

Disk Management shows two basic disks: Disk 0 (223.57 GB, MBR, data) and Disk 1 (238.46 GB, GPT, system). Disk 0 contains the 50 MB “Réservé au système” partition, the 125.36 GB F: volume, a 519 MB recovery partition, and an extended partition holding the 97.65 GB E: logical drive. Disk 1 contains the 100 MB EFI System Partition, a 16 MB Microsoft Reserved partition, the 147.42 GB C: system volume and the 90.94 GB G: data volume. The 1 MB and 2 MB gaps visible before some partitions are alignment reserved areas, not lost data.

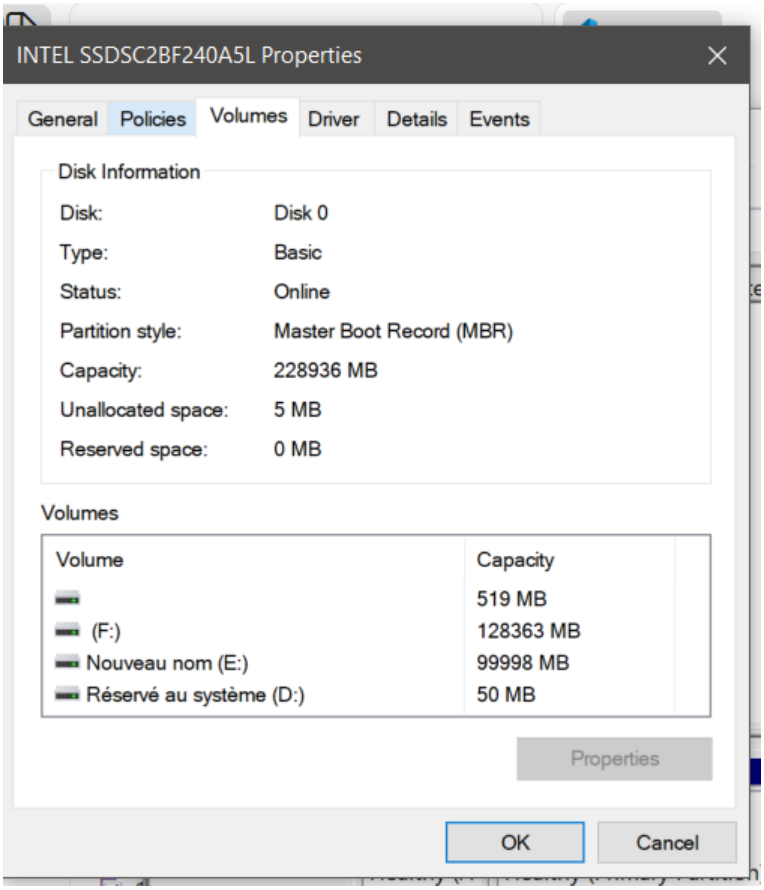


Figure 7 – Disk 0 Properties, Volumes tab: Partition style = Master Boot Record (MBR) (screenshot 7).

The Volumes tab of Disk 0 confirms its partition style is Master Boot Record. This is an important detail: the machine boots in UEFI mode, but its boot disk is Disk 1 (GPT); Disk 0 is an MBR data disk. UEFI firmware can still read MBR data disks, so the two partitioning schemes coexist without conflict.

6.2 PowerShell inspection (D1)

```
PS C:\Windows\system32> Get-Disk

Number Friendly Name Serial Number                  HealthStatus         OperationalStatus      Total Size Partition
                                                                                                            Style
------ ------------- -------------                  ------------         -----------------      ---------- ----------
0      INTEL SSDS... CVTS5403057M240JGN             Healthy              Online                  223.57 GB MBR
1      PC611 NVMe... ACE4_2E00_0524_9834_2EE4_AC00... Healthy            Online                  238.47 GB GPT


PS C:\Windows\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_intel&prod_ssdsc2bf240a51#4&f78d1d6&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                D           1048576                                     50 MB IFS
2                F           53477376                                   125.36 GB IFS
0                           134652887040                                97.66 GB XINT13 Extended
4                E           134653935616                                97.65 GB IFS
3                           239510487040                                 519 MB Unknown


   DiskPath: \\?\scsi#disk&ven_nvme&prod_pc611_nvme_sk_hy#5&1bfa0ee4&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                  147.42 GB Basic
4                G           158410825728                                90.94 GB Basic


PS C:\Windows\system32> Get-Volume

DriveLetter FriendlyName       FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------       -------------- --------- ------------ ----------------- -------------      ----
                               FAT32          Fixed     Healthy      OK                    69.11 MB     96 MB
C                              NTFS           Fixed     Healthy      OK                    11.78 GB 147.42 GB
G           New Volume         NTFS           Fixed     Healthy      OK                     6.89 GB  90.94 GB
E           Nouveau nom        NTFS           Fixed     Healthy      OK                    14.28 GB  97.65 GB
D           Réservé au système NTFS           Fixed     Healthy      OK                     2.06 MB     50 MB
F                              NTFS           Fixed     Healthy      OK                     8.17 GB 125.36 GB
                               NTFS           Fixed     Healthy      OK                    84.71 MB    519 MB
```

Figure 8 – Get-Disk, Get-Partition and Get-Volume output (screenshot 8).

Get-Disk reports PartitionStyle = MBR for Disk 0 and GPT for Disk 1, with LogicalSectorSize = 512 on both disks (Disk 1 reports a 4096-byte physical sector size, which is normal for NVMe media). Get-Partition provides the byte offsets used later to derive start sectors, and Get-Volume provides the file system and volume sizes.

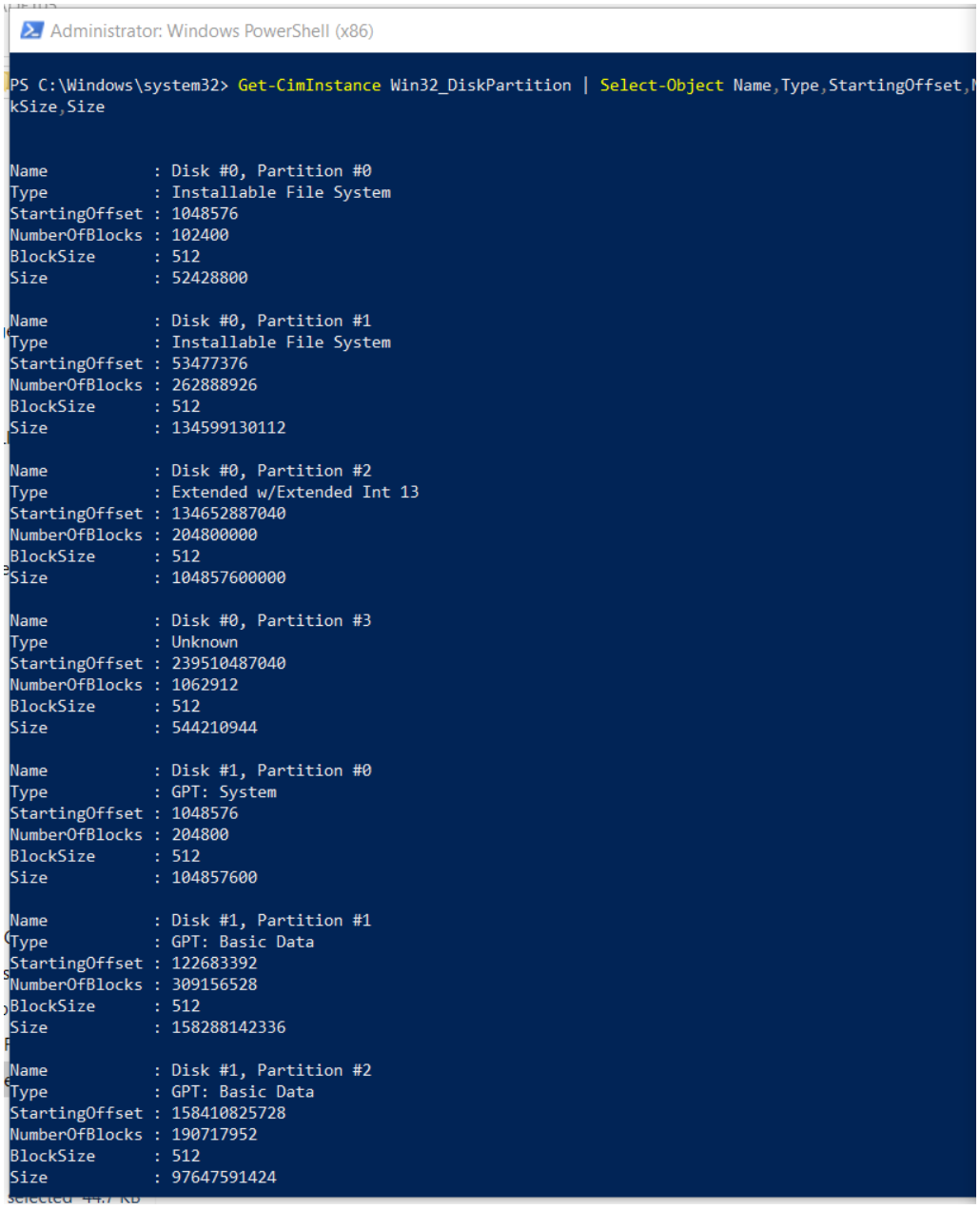


Figure 9 – Win32_DiskPartition output from an elevated PowerShell window: offsets, block counts and block size (screenshot 9).

The CIM query is the key source for the final assignment because it exposes NumberOfBlocks (the sector count) and BlockSize (bytes per sector) directly. Win32_DiskPartition numbers partitions from zero, so “Disk #0, Partition #0” is Partition 1 in DiskPart and Get-Partition. This zero-based versus one-based difference must be stated explicitly to avoid misidentifying a partition.

6.3 DiskPart listing (D1)

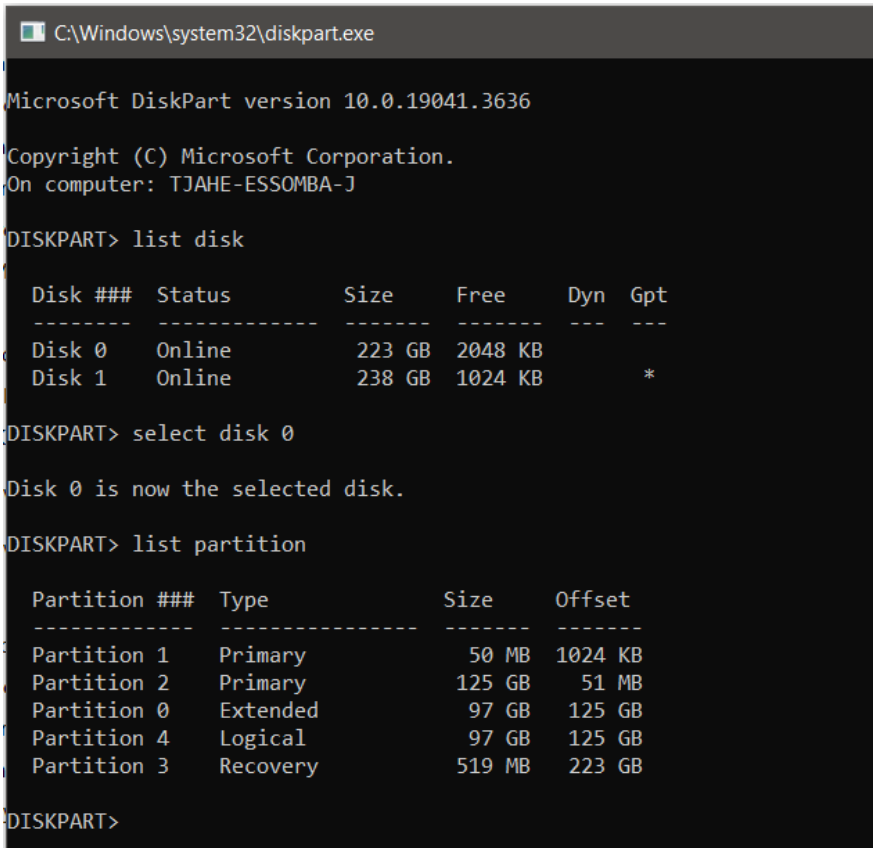


Figure 10 – DiskPart: list disk, select disk 0, list partition (listing only, no modifications) (screenshot 10).

DiskPart was opened in an administrator Command Prompt and used only for listing. It shows Disk 0 (223 GB, MBR) and Disk 1 (238 GB, GPT), and for Disk 0 it lists Partition 1 (Primary), Partition 2 (Primary), Partition 0 (Extended), Partition 4 (Logical) and Partition 3 (Recovery). DiskPart reports offsets in KB, so the 1,024 KB offset of Partition 1 corresponds to byte offset 1,048,576 and start sector 2,048. The unusual order of the partition numbers is simply the order of entries in the MBR partition table, not a sign of damage.

6.4 Full partition layout derived from the evidence

Disk	Partition	Offset (bytes)	Start sector	Sectors	Size	Type / FS
0 (MBR)	1 (D:)	1,048,576	2,048	102,400	50 MB	Primary / NTFS
0 (MBR)	2 (F:)	53,477,376	104,448	262,888,926	125.36 GB	Primary / NTFS
0 (MBR)	0 (container)	134,652,887,040	262,994,232	204,800,000	97.66 GB	Extended (XINT13)
0 (MBR)	4 (E:)	134,653,935,616	262,995,968	204,795,904	97.65 GB	Logical / NTFS
0 (MBR)	3 (recovery)	239,510,487,040	467,793,920	1,062,912	519 MB	Recovery / NTFS
1 (GPT)	1 (ESP)	1,048,576	2,048	204,800	100 MB	EFI System / FAT32
1 (GPT)	2 (MSR)	105,906,176	206,848	32,768	16 MB	Microsoft Reserved
1 (GPT)	3 (C:)	122,683,392	239,616	309,156,528	147.42 GB	Basic Data / NTFS
1 (GPT)	4 (G:)	158,410,825,728	309,396,144	190,717,952	90.94 GB	Basic Data / NTFS

Start sectors were obtained by dividing each StartingOffset by the 512-byte block size, and sector counts are the NumberOfBlocks values. The table reconciles three independent tools: Disk Management (sizes), PowerShell (offsets and volumes) and DiskPart (partition numbering and KB offsets).

6.5 Linux and WinHex alternatives (D2–D3)

The Windows path was selected for this lab, so the Linux commands (lsblk, fdisk -l, parted -l, blkid, df -h) were not executed; a Kali VM is available on the machine but the required evidence was already obtainable

natively. The optional WinHex step was also skipped: on a UEFI/GPT disk, sector 0 would contain a protective MBR with a single type-EE entry and the 55 AA signature, while the real GPT header sits in LBA 1. Since the optional step requires instructor permission and a forensic viewer, it was not performed; the GPT information was instead verified with Get-Disk and Disk Management.

7. Part E – Boot process, MBR and GPT

7.1 Boot sequence of this PC (E1)

The boot process of the examined machine follows the UEFI path below. Each stage is a potential source of forensic artefacts.

```
1. Power-on and POST
   Firmware initialises the CPU, memory and devices and performs the power-on self-test.

2. UEFI firmware and NVRAM settings
   Boot order, Secure Boot and other firmware settings are read from NVRAM (the modern
   equivalent of CMOS settings).

3. Boot device selection and EFI boot
   The firmware reads the GPT, finds the EFI System Partition (Disk 1, Partition 1, FAT32,
   100 MB) and loads the Windows Boot Manager (\EFI\Microsoft\Boot\bootmgfw.efi).

4. Boot manager
   Windows Boot Manager reads the BCD store, offers a boot menu if several entries exist,
   and loads winload.efi.

5. Kernel loading
   winload.efi loads ntoskrnl.exe, the HAL and boot-start drivers; with Secure Boot the
   images are signature-checked before execution.

6. Kernel initialisation
   The kernel starts smss.exe, then wininit.exe, services.exe, lsass.exe and winlogon.exe,
   and mounts the system volume.

7. Operating system control
   The user session starts (explorer.exe), services run, and logging and user activity begin.
```

Stage	What happens	Forensic relevance
Power-on and POST	Hardware is initialised and tested.	Hardware faults or tampering can affect access to evidence.
UEFI/BIOS firmware	Boot order and firmware settings are applied from NVRAM/CMOS.	Boot mode explains the partition style and the boot artefacts to expect.
GPT/ESP or MBR boot	Firmware locates the EFI boot entry or reads the MBR boot code.	Boot records can reveal boot configuration and tampering.
Boot manager	Windows Boot Manager reads BCD and selects the OS entry.	May show multi-boot setups or a hidden second OS.
Kernel and OS control	Kernel and drivers start; user session begins.	Event logs, prefetch and user activity start accumulating.

Evidence for the boot mode: msinfo32 reports BIOS Mode = UEFI (Figure 1), Get-Disk reports the system/boot disk as GPT with IsSystem = True and IsBoot = True (Figure 8), and msinfo32 reports the boot device as \Device\HarddiskVolume5. The combination UEFI + GPT + EFI System Partition is internally consistent. The secondary SATA disk being MBR does not affect booting because it contains no boot entry; it is a data disk.

7.2 MBR compared with GPT (E2)

Feature	MBR	GPT
Age and design	Introduced in the 1980s; legacy BIOS era.	Modern standard designed for UEFI systems.
Maximum disk size	Commonly limited to about 2 TB.	Supports very large disks (slide: up to about 9.4 ZB).
Partition information	Partition table and boot code share sector 0.	Primary GPT header at LBA 1 plus a backup header at the end of the disk.

Feature	MBR	GPT
Number of partitions	4 primary entries; more only through an extended/logical chain.	128 entries by default in Windows.
Redundancy	Single copy of the partition table; sector 0 damage is critical.	Backup header and entries allow recovery and validation.
Forensic issue	Damage or malware in sector 0 can make partitions unreadable.	Two copies must be compared; a mismatch can indicate tampering.
Seen on this PC	Disk 0 (data disk) uses MBR.	Disk 1 (system/boot disk) uses GPT.

The coexistence of the two schemes on one machine is a useful forensic lesson: the boot mode applies to the boot disk, not to every disk in the system. During acquisition, the MBR of Disk 0 and the GPT headers of Disk 1 must both be preserved, because each carries different structural evidence. The GPT backup header at the end of Disk 1 is particularly valuable if the primary header is damaged.

8. Part F – File system identification

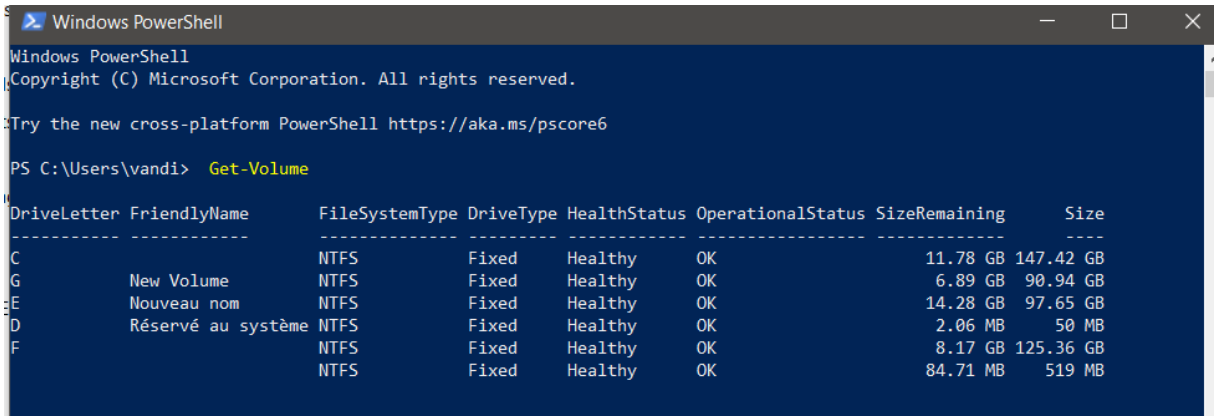


Figure 11 – Get-Volume output showing the file system of every mounted volume (screenshot 11).

Two file systems are present on the machine: NTFS on all Windows data volumes and FAT32 on the EFI System Partition. The Microsoft Reserved partition carries no file system. This matches the expected layout of a UEFI Windows installation.

Volume / partition	File system	Size	Interpretation
Disk 1, Partition 1 (ESP)	FAT32	100 MB	UEFI firmware must read this partition; FAT32 is the required ESP format.
Disk 1, Partition 2 (MSR)	None (reserved)	16 MB	Reserved by Windows for future metadata; not user-accessible.
Disk 1, Partition 3 (C:)	NTFS	147.42 GB	Windows system volume: MFT, USN journal, event logs and user profiles.
Disk 1, Partition 4 (G:)	NTFS	90.94 GB	Data volume with standard NTFS metadata and permissions.
Disk 0, Partition 1 (D:)	NTFS	50 MB	Small system-reserved NTFS volume created by the installation.
Disk 0, Partition 2 (F:)	NTFS	125.36 GB	Data volume.
Disk 0, logical drive (E:)	NTFS	97.65 GB	Logical drive inside the MBR extended partition.
Disk 0, Partition 3	NTFS	519 MB	Recovery volume (Windows RE).

File system	Common OS / usage	Forensic importance	Seen on this PC?
FAT/FAT32	Removable media and older systems; EFI System Partition.	Simple structure with a file allocation table; commonly used on USB drives and required for the ESP.	Yes – ESP (100 MB)
NTFS	Modern Windows systems.	MFT entries, standard timestamps, permissions, alternate data streams and $LogFile journal enable timeline and recovery analysis.	Yes – C:, D:, E:, F:, G:, recovery
ext family (ext2/3/4)	Linux systems.	Inode-based metadata, journaling and timestamps	No

File system	Common OS / usage	Forensic importance	Seen on this PC?
		support timeline analysis.	
XFS / JFS / btrfs	Linux and Unix-like systems.	Advanced allocation and copy-on-write features affect recovery and metadata interpretation.	No
HFS+ / APFS	Older and current macOS systems.	Important in Mac investigations; HFS+ has documented date-range limitations.	No

File systems matter in digital forensics because they determine how data and metadata are stored, and therefore what can be recovered and how it can be interpreted. NTFS keeps a record of every file in the Master File Table, including timestamps (creation, modification, access, MFT change), permissions and alternate data streams; its $LogFile and USN journal can reveal recent changes. The file system also defines cluster size, which determines how much slack space exists at the end of the last cluster of a file, and it determines whether deleted content can be carved or recovered. Different file systems require different tools and different interpretation rules, so identifying the file system is always an early step in an investigation. FAT32 on the ESP is also relevant: it is small, simple and can be modified by an attacker to influence booting.

9. Part G – Final practical assignment: verification of partitions 1 and 2

The assignment requires the size of Partitions 1 and 2 to be verified from the partition table, not from the total disk size. The two partitions analysed are the first two primary partitions of Disk 0, which DiskPart and Get-Partition both number 1 and 2. The evidence comes from four independent read-only sources: DiskPart (Figure 10), Win32_DiskPartition (Figures 9 and 12), Get-Partition/Get-Volume (Figures 8 and 11) and Disk Management (Figure 6).

9.1 Verification process, step by step

1. Disk Management was opened and Disk 0 was identified as the 223.57 GB MBR disk. Partition 1 is the 50 MB “Réservé au système” volume (D:) and Partition 2 is the 125.36 GB volume (F:) (Figure 6).

2. DiskPart was used in listing mode. list disk showed Disk 0 as 223 GB MBR; select disk 0 selected it without modifying it; list partition showed Partition 1 (Primary), Partition 2 (Primary), Partition 0 (Extended), Partition 4 (Logical) and Partition 3 (Recovery) (Figure 10).

3. Get-Partition was run to obtain the byte offsets: 1,048,576 for Partition 1 and 53,477,376 for Partition 2 (Figure 8).

4. Win32_DiskPartition was queried in an elevated PowerShell window to obtain NumberOfBlocks and BlockSize. For Disk #0, Partition #0 (Partition 1 in DiskPart) the sector count is 102,400; for Disk #0, Partition #1 (Partition 2 in DiskPart) it is 262,888,926. BlockSize is 512 in both cases (Figures 9 and 12).

5. Start sectors were derived by dividing the offsets by 512: 1,048,576 ÷ 512 = 2,048 and 53,477,376 ÷ 512 = 104,448. End sectors were calculated as start + count − 1.

6. The size in bytes, KB, MB and GB was calculated for each partition and compared with the OS-reported size (Disk Management and Get-Volume). The small differences were then explained.

9.2 Partition 1 – identification and sector data

```
PS C:\Users\vandi> Get-CimInstance Win32_DiskPartition | Select-Object Name,Type,StartingOffset,NumberOfBlocks,BlockSize,Size


Name           : Disk #0, Partition #0
Type           : Installable File System
StartingOffset : 1048576
NumberOfBlocks : 102400
BlockSize      : 512
Size           : 52428800

Name           : Disk #0, Partition #1
Type           : Installable File System
StartingOffset : 53477376
NumberOfBlocks : 262888926
BlockSize      : 512
Size           : 134599130112

Name           : Disk #0, Partition #2
Type           : Extended w/Extended Int 13
StartingOffset : 134652887040
NumberOfBlocks : 204800000
BlockSize      : 512
Size           : 104857600000

Name           : Disk #0, Partition #3
Type           : Unknown
StartingOffset : 239510487040
NumberOfBlocks : 1062912
BlockSize      : 512
Size           : 544210944

Name           : Disk #1, Partition #0
Type           : GPT: System
StartingOffset : 1048576
NumberOfBlocks : 204800
BlockSize      : 512
Size           : 104857600

Name           : Disk #1, Partition #1
Type           : GPT: Basic Data
StartingOffset : 122683392
NumberOfBlocks : 309156528
BlockSize      : 512
Size           : 158288142336

Name           : Disk #1, Partition #2
Type           : GPT: Basic Data
StartingOffset : 158410825728
NumberOfBlocks : 190717952
BlockSize      : 512
Size           : 97647591424
```

Figure 12 – Win32_DiskPartition evidence used for Partition 1 and Partition 2 (screenshot 12; same query as Figure 9).

Item	Value / evidence
Partition identity	Disk 0, Partition 1 (DiskPart / Get-Partition); “Disk #0, Partition #0” in WMI
Drive letter and file system	D:, NTFS (Get-Volume, Figure 11)
Partition type	Primary (DiskPart, Figure 10)
Starting offset	1,048,576 bytes (Get-Partition, Figure 8)
Start sector	1,048,576 ÷ 512 = 2,048
Sector count	102,400 (NumberOfBlocks, Figure 9)
End sector	2,048 + 102,400 − 1 = 104,447
Bytes per sector	512 (BlockSize)
Raw size	102,400 × 512 = 52,428,800 bytes

9.3 Partition 1 – calculation in GB

```
Partition 1 (Disk 0, Partition 1 – D:)
  Sector count        = 102,400
  Bytes per sector    = 512
  Size in bytes       = 102,400 × 512            = 52,428,800 bytes
  Size in KB          = 52,428,800 ÷ 1,024       = 51,200 KB
  Size in MB          = 51,200 ÷ 1,024           = 50 MB
```

```
  Size in GB (GiB)    = 50 ÷ 1,024              = 0.048828125 GB ≈ 0.0488 GB
  Decimal GB          = 52,428,800 ÷ 1,000,000,000 = 0.0524 GB
```

Verification in Python: 102400 * 512 / 1024**3 = 0.048828125. The OS reports this volume as 50 MB (Disk Management, Figure 6; Get-Volume, Figure 11), which is exactly 0.0488 GiB rounded to a whole number of MB.

9.4 Partition 2 – identification and sector data

Item	Value / evidence
Partition identity	Disk 0, Partition 2 (DiskPart / Get-Partition); “Disk #0, Partition #1” in WMI
Drive letter and file system	F:, NTFS (Get-Volume, Figure 11)
Partition type	Primary (DiskPart, Figure 10)
Starting offset	53,477,376 bytes (Get-Partition, Figure 8)
Start sector	53,477,376 ÷ 512 = 104,448
Sector count	262,888,926 (NumberOfBlocks, Figure 9)
End sector	104,448 + 262,888,926 − 1 = 262,993,373
Bytes per sector	512 (BlockSize)
Raw size	262,888,926 × 512 = 134,599,130,112 bytes

9.5 Partition 2 – calculation in GB

```
Partition 2 (Disk 0, Partition 2 – F:)
  Sector count        = 262,888,926
  Bytes per sector    = 512
  Size in bytes       = 262,888,926 × 512          = 134,599,130,112 bytes
  Size in KB          = 134,599,130,112 ÷ 1,024    = 131,444,463 KB
  Size in MB          = 131,444,463 ÷ 1,024        = 128,363.733 MB
  Size in GB (GiB)    = 128,363.733 ÷ 1,024        = 125.3552 GB ≈ 125.36 GB
  Decimal GB          = 134,599,130,112 ÷ 1,000,000,000 = 134.599 GB

>>> 262888926 * 512 / 1024**3
125.35520839691162
```

The OS reports this volume as 125.36 GB in both Disk Management (Figure 6) and Get-Volume (Figure 11), which matches the calculated 125.3552 GiB after rounding to two decimals.

9.6 Comparison with the OS-reported size

Partition	Calculated raw size	Calculated (binary GB/GiB)	OS-reported size	Difference
1 (D:)	52,428,800 bytes	0.0488 GiB (50 MB)	50 MB (Disk Management; Get-Volume)	0 – identical after rounding
2 (F:)	134,599,130,112 bytes	125.3552 GiB ≈ 125.36 GB	125.36 GB (Disk Management; Get-Volume)	0.00 GB after rounding

A second, more precise comparison uses the byte values reported by WMIC logicaldisk. For Partition 1, WMIC reports 52,424,704 bytes against the partition-table value of 52,428,800 bytes, a difference of 4,096 bytes (4 KiB). For Partition 2, WMIC reports 134,599,127,040 bytes against 134,599,130,112 bytes, a difference of 3,072 bytes (3 KiB). These are not calculation errors: the partition table describes the raw partition, while the volume size reported by the file system excludes a small amount of space used by NTFS metadata and the backup boot sector at the end of the volume.

9.7 Explanation of the differences

Three effects explain every difference observed. First, Windows uses binary units: it divides by 1,024 and still labels the result “GB”, so 134,599,130,112 bytes become 125.36 GiB, while the decimal value is 134.599 GB. This is the same effect that makes a “256 GB” NVMe drive appear as 238.47 GB in Windows. Second, tools round to different numbers of decimals, so Disk Management, Get-Volume and a calculator may display

125.36, 125.35 or 125.4 for the same partition. Third, file-system overhead: NTFS reserves metadata and a backup boot sector, which is why the volume size is a few kilobytes smaller than the raw partition. Finally, the first partition starts at sector 2,048 (1 MiB) rather than sector 0 because modern partitioning tools align partitions to 1 MiB boundaries; the gap contains the MBR and alignment padding and is not usable storage.

As an additional cross-check, the same method was applied to the GPT boot disk: Disk 1, Partition 3 (C:) has 309,156,528 sectors × 512 = 158,288,142,336 bytes = 147.42 GiB, and Disk 1, Partition 4 (G:) has 190,717,952 sectors × 512 = 97,647,591,424 bytes = 90.94 GiB. Both match the sizes shown in Disk Management, which confirms that the method is reliable across both MBR and GPT disks.

10. Conclusion

This lab produced a complete, read-only profile of the workstation and demonstrated how much forensic information can be obtained without altering a single sector. The machine is a UEFI-based HP ZBook 15u G3 running Windows 10 Pro on a GPT NVMe boot disk, with a secondary MBR SATA disk. The system layers, the running processes, the partition table and the file systems were all documented from independent tools, and the partition sizes were calculated manually from sector counts and then verified against the operating system. Partition 1 (50 MB) and Partition 2 (125.36 GB) matched the OS-reported sizes after accounting for binary units, rounding and a few kilobytes of NTFS overhead.

The exercise also showed why forensic work must be methodical. A single action leaves traces in the user, application, operating-system, disk and memory layers, and the same fact can be verified from several of them. Process and memory data are volatile and must be captured live, while partition tables and file-system metadata persist and can be re-examined later. Understanding sector arithmetic, MBR/GPT differences and file-system behaviour is therefore not academic: it determines whether an investigator can correctly interpret a disk, recover deleted data and defend the findings in a report or in court. The discipline of using only listing and query commands, recording screenshots, showing calculations and explaining every difference is the core habit that this lab set out to build.

11. References

ICDFA. (2026). Computer System Introduction [Course slide deck]. CIP-B101 Basic Computer Skills for Digital Forensics.

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Microsoft. (n.d.). DiskPart command-line options. Microsoft Learn. https://learn.microsoft.com/windows-server/administration/windows-commands/diskpart

NIST. (2006). Guide to Integrating Forensic Techniques into Incident Response (SP 800-86). National Institute of Standards and Technology.

Appendix A – Screenshot checklist

No.	Required screenshot	Included	Figure
1	System Information or systeminfo output	Yes	Figures 1 and 2
2	Disk drive model and size output	Yes	Figure 3
3	Logical volume / file system output	Yes	Figures 3, 8 and 11
4	Running process / PID output	Yes	Figures 4 and 5
5	Disk Management, Get-Partition, DiskPart, lsblk, fdisk or parted output	Yes	Figures 6, 8 and 10
6	Partition 1 calculation evidence	Yes	Figures 9, 10 and 12
7	Partition 2 calculation evidence	Yes	Figures 9, 10 and 12
8	Boot mode evidence: BIOS/UEFI	Yes	Figure 1 (BIOS Mode = UEFI)
9	File system identification evidence	Yes	Figures 3 and 11
10	Final report summary / conclusion page	Yes	Section 10

Appendix B – Raw command output used in the report

The text below reproduces the relevant parts of the PowerShell output so that the figures can be cross-checked without re-running the commands.

```
PS C:\> Get-Disk
Number  Friendly Name            Size      Partition Style
0       INTEL SSDSC2BF240A5L      223.57 GB MBR
1       PC611 NVMe SK hynix 256GB 238.47 GB GPT

PS C:\> Get-Partition
Disk 0  Partition 1  D:  Offset 1048576      Size 52428800
Disk 0  Partition 2  F:  Offset 53477376     Size 134599130112
Disk 0  Partition 0      Offset 134652887040 Size 104857600000  (Extended)
Disk 0  Partition 4  E:  Offset 134653935616 Size 104855502848
Disk 0  Partition 3      Offset 239510487040 Size 544210944     (Recovery)
Disk 1  Partition 1      Offset 1048576      Size 104857600     (EFI System)
Disk 1  Partition 2      Offset 105906176    Size 16777216      (MSR)
Disk 1  Partition 3  C:  Offset 122683392    Size 158288142336
Disk 1  Partition 4  G:  Offset 158410825728 Size 97647591424

PS C:\> Get-CimInstance Win32_DiskPartition |
         Select-Object Name,Type,StartingOffset,NumberOfBlocks,BlockSize,Size

Name                 Type                          StartingOffset   NumberOfBlocks  BlockSize
Size
Disk #0, Partition #0 Installable File System       1048576         102400          512
52428800
Disk #0, Partition #1 Installable File System       53477376        262888926       512
134599130112
Disk #0, Partition #2 Extended w/ Extended Int 13   134652887040    204800000       512
104857600000
Disk #0, Partition #3 Unknown                       239510487040    1062912         512
544210944
Disk #1, Partition #0 GPT: System                   1048576         204800          512
104857600
Disk #1, Partition #1 GPT: Basic Data               122683392       309156528       512
158288142336
Disk #1, Partition #2 GPT: Basic Data               158410825728    190717952       512
97647591424
```